

A Period-Three Limit Orbit in a Thue–Morse Heteroclinic Closure

Route-A structural certificate C154

Abstract

We join a period-three past to a Thue–Morse future in one two-sided symbolic configuration and close its full shift orbit. The closure has exactly the Thue–Morse component, the interface orbit, and the three periodic phases. Consequently its only primitive orbit has length three and its Artin–Mazur zeta is $(1 - z^3)^{-1}$. This periodic layer is generated by one heteroclinic closure rather than freely attached.

1 The frozen interface

Let t_n be binary digit-sum parity for $n \geq 0$, and let $y_j = (2, 3, 4)_{j \bmod 3}$. Define

$$x_j = \begin{cases} y_j, & j < 0, \\ t_j, & j \geq 0, \end{cases} \quad X = \overline{\{\sigma^n x : n \in \mathbb{Z}\}},$$

where $(\sigma x)_j = x_{j+1}$. If $n_k \rightarrow +\infty$, every fixed window of $\sigma^{n_k} x$ eventually lies in the binary tail, so each limit lies in X_{TM} . Conversely, uniform recurrence puts every finite word of every X_{TM} point arbitrarily far out in t ; nested occurrences and a diagonal subsequence realize every such point as a positive-shift limit. If $n_k \rightarrow -\infty$, every fixed window lies eventually in the periodic tail; its residue class modulo three gives one phase of y , and all phases occur. A bounded convergent integer-shift sequence is eventually constant, while an unbounded sequence has a subsequence escaping in one direction. Thus

$$X = X_{\text{TM}} \sqcup \text{Orb}(x) \sqcup \text{Orb}(y). \quad (1)$$

The pair 40 occurs exactly once in $\sigma^n x$, at coordinates $-n - 1, -n$, and occurs in neither limit component. Its cylinder is therefore the singleton $\{\sigma^n x\}$, so every interface point is isolated and wandering. The full orbit of x is dense in X (a two-sided transitivity statement), while the period-three orbit is a proper closed invariant set; hence X is not minimal.

2 Periodic orbit ledger

The Thue–Morse language has no periodic point, and the isolated wandering interface orbit has none. The two pure components are nonwandering and the interface complement is wandering, so $\Omega(\sigma) = X_{\text{TM}} \sqcup \text{Orb}(y)$. Therefore

$$\# \text{Fix}(\sigma^n) = \begin{cases} 3, & 3 \mid n, \\ 0, & 3 \nmid n. \end{cases} \quad (2)$$

Möbius inversion leaves three exact-period points at $n = 3$, hence one primitive cycle. Moreover

$$\log \zeta_X(z) = \sum_{q \geq 1} \frac{3z^{3q}}{3q} = -\log(1 - z^3), \quad \zeta_X(z) = \frac{1}{1 - z^3}. \quad (3)$$

The exact ledgers are implementation sentinels, not theorem cutoffs. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL). We claim no target divisor, arithmetic/local factor, target functional equation or counting law, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.